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**Review**

Review

# Genetically guided precision medicine clinical decision support tools: a systematic review

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## Abstract

**Objectives:** Patient care using genetics presents complex challenges. Clinical decision support (CDS) tools are a potential solution because they provide patient-specific risk assessments and/or recommendations at the point of care. This systematic review evaluated the literature on CDS systems which have been implemented to support genetically guided precision medicine (GPM).

**Materials and Methods:** A comprehensive search was conducted in MEDLINE and Embase, encompassing January 1, 2011–March 14, 2023. The review included primary English peer-reviewed research articles studying humans, focused on the use of computers to guide clinical decision-making and delivering genetically guided, patient-specific assessments, and/or recommendations to healthcare providers and/or patients.

**Results:** The search yielded 3832 unique articles. After screening, 41 articles were identified that met the inclusion criteria. Alerts and reminders were the most common form of CDS used. About 27 systems were integrated with the electronic health record; 2 of those used standards-based approaches for genomic data transfer. Three studies used a framework to analyze the implementation strategy.

**Discussion:** Findings include limited use of standards-based approaches for genomic data transfer, system evaluations that do not employ formal frameworks, and inconsistencies in the methodologies used to assess genetic CDS systems and their impact on patient outcomes.

**Conclusion:** We recommend that future research on CDS system implementation for genetically GPM should focus on implementing more CDS systems, utilization of standards-based approaches, user-centered design, exploration of alternative forms of CDS interventions, and use of formal frameworks to systematically evaluate genetic CDS systems and their effects on patient care.

**Key words:** clinical decision support systems; genomics; genetics; electronic health record; personalized medicine.

## Background

Genetic disorders, while individually rare, are collectively common.[1] These disorders affect between 2% and 10% of the population, occur in all medical specialties, and are more common in specific populations.[2–5] Genetically guided precision medicine (GPM) entails the delivery of individually tailored medical care that leverages information about each person's unique characteristics, including clinical data, genetic test results, patient preferences, and family health history (FHx).[6–8] GPM is expanding with advances in genomic tools and decreasing genetic testing costs.[9–11] Genomic testing can facilitate diagnosis and inform condition-specific clinical care.[12–15] However, evidence indicates that providers have limited proficiency, self-efficacy, and resources to guide decisions about when and how to order genetic tests, refer to specialists, or change treatment or surveillance based on genomic information.[16–18] As a result, genomic testing in clinical care is underutilized.[8,19,20] The challenges facing clinicians will continue to grow as the volume of information generated expands the evidence base for linking genetic variation with human disease.[21] For GPM to achieve the potential to tailor medical treatments and therapies to the individual characteristics of each patient, new approaches are needed to support the integration of genomic medicine in clinical decision-making, especially given the limited genomic specialist workforce.[22]

The essential role of health information technology in overcoming the barriers that GPM faces is recognized.[23–25] A key challenge is the integration of genomic data into electronic health records (EHRs), as genomics data possess unique characteristics that set them apart from other types of clinical data. Genomic data are highly complex, voluminous, and dynamic, yet germline genetic test results do not change over an individual's lifetime. These attributes require specialized approaches for storage, retrieval, and ongoing interpretation.[26,27] Currently, genomic data are often poorly integrated into EHRs, resulting in suboptimal clinical use.

Clinical decision support (CDS) systems have been proposed to address these barriers by facilitating the integration

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and utilization of data, including genomic data in clinical care.[28] CDS systems may support clinicians in ordering genetic testing and the management of care, preventing harmful or unnecessary interventions, and decreasing delays in diagnosis, which can lead to excessive healthcare utilization and potentially inappropriate testing and treatment.[29] CDS systems can present information to users in a variety of ways, such as standalone systems (either electronic or paper) or systems which are integrated into the clinical workflow via the EHR.[30] CDS systems may be used asynchronously, meaning use would not be associated with a specific patient encounter, instead identifying individuals in need of a given service (eg, vaccination, mammogram) as part of population health initiatives. CDS systems can also be used synchronously when presenting patient-specific information within the context of an encounter. By guiding assessments or recommendations at the point of care based on clinical management guidelines, best practices, and evidence,[31] CDS systems have the potential to effectively influence changes in clinical care, thus facilitating the use of genomic data in clinical decision-making.[32]

In 2011, Welch and Kawamoto conducted a systematic review about genetic CDS systems, including prototypes and EHR-integrated CDS systems.[33] This review showed an abundance of CDS systems in prototype stage and stand-alone CDS systems, with very few implemented outside of a pilot setting. Since this review, the use and evaluation of genetic CDS systems in clinical settings have proliferated, yet there has not been a review of studies summarizing the implementation, use, and evaluation of these in health care settings. To understand the evolution of the development and implementation of genomic CDS systems, we conducted a systematic review of genomic CDS systems which have been implemented in a clinical setting.

## Methods

A systematic review was conducted to integrate quantitative and qualitative evidence using the Rapid and Rigorous Qualitative Data Analysis technique[34] to examine elements surrounding implemented genomic clinical decision support (gCDS) tools.

### Literature search strategy

A bioinformaticist (D.J.) consulted with a PhD researcher/medical librarian (K.M.R.) to search MEDLINE and Embase from January 1, 2011 to March 14, 2023, using a search strategy adapted from previous systematic reviews of CDS,[33] genetic testing, genetic health services, and family history (FHx) (Appendix S1). The initial literature search was conducted on November 15, 2021, and an additional literature search was conducted on March 14, 2023, to capture additional citations. Both subject headings and text terms were used to search for CDS tools and genetic testing, including FHx. Reference lists of included articles and relevant systematic reviews were hand-searched for additional references.

### Study selection

References were reviewed against the following inclusion criteria: primary English peer-reviewed research article studying humans, focused on the use of computers to guide clinical decision-making and delivering genetically guided, patient-specific assessments and/or recommendations to healthcare providers and/or patients. If there were any genetic specific data that were used or presented by a CDS, the article would be included provided other inclusion criteria were met. Two independent reviewers (D.J. and M.S.W./N.S./E.J./E.I./C.J./J. C.) conducted literature screening using DistillerSR and a screening form. Titles and abstracts were screened to assess whether the articles met inclusion criteria. Conflicting decisions were discussed between the 2 reviewers, with disagreements resolved through discussion with a third reviewer (M.S.W.) as a final decision maker. A similar process was followed for full-text reviews. Each of the research papers selected met all the inclusion criteria.

### Data abstraction

Quantitative and qualitative data extraction were completed by D.J. on each of the articles that met the inclusion criteria. Any data points that could not be determined were marked as unknown or not reported. A paper could potentially meet multiple criteria but we labeled each article by its primary directive and categorized under that data element. Annotation was completed for each paper, rather than for each gCDS tool. Abstraction categorized each paper under the data elements. Data table definitions can be found in Appendix S2. Data extraction items for all articles are described in Appendix S3.

### Data analysis

Using the abstracted attributes, the manuscripts were grouped into categories according to CDS type and clinical application area. The findings from these manuscripts were summarized through tables and narrative discussion. Notable themes and trends were identified and discussed. Reported barriers and facilitators were noted. Each article was classified according to the CDS taxonomy[35] and CDS critical success features.[36] A quantitative analysis of CDS trials to identify features critical to successful CDS tools was considered. However, due to the limited sample size of CDS trials and the lack of implementation data, this analysis was not feasible. A graph to show the trends of publications was created.

## Results

The initial MEDLINE and Embase searches identified 3832 unique potentially relevant articles. The title and abstract review excluded 3438 articles. The remaining 394 underwent full-text review, in which 353 were rejected. The PRISMA diagram (Figure 1) articulates the inclusion process. This left 41 primary research articles for analysis that were published from January 1, 2011 to March 14, 2023 (Table 1).[37–77] Of those 41 primary research articles, there were 36 unique gCDS tools implemented.

The described CDS tools were first categorized according to 3 areas of clinical focus: genetically guided cancer management, pharmacogenomics (PGx), and other systems for precision medicine. CDS tools were further categorized by type of software architecture approach.

- *Stand-alone CDS*: A stand-alone CDS tool which utilizes information outside of the EHR and does not interact or exchange data with the EHR (14 articles).
- *EHR proprietary*: A CDS tool that works within the native EHR but does not report using industry

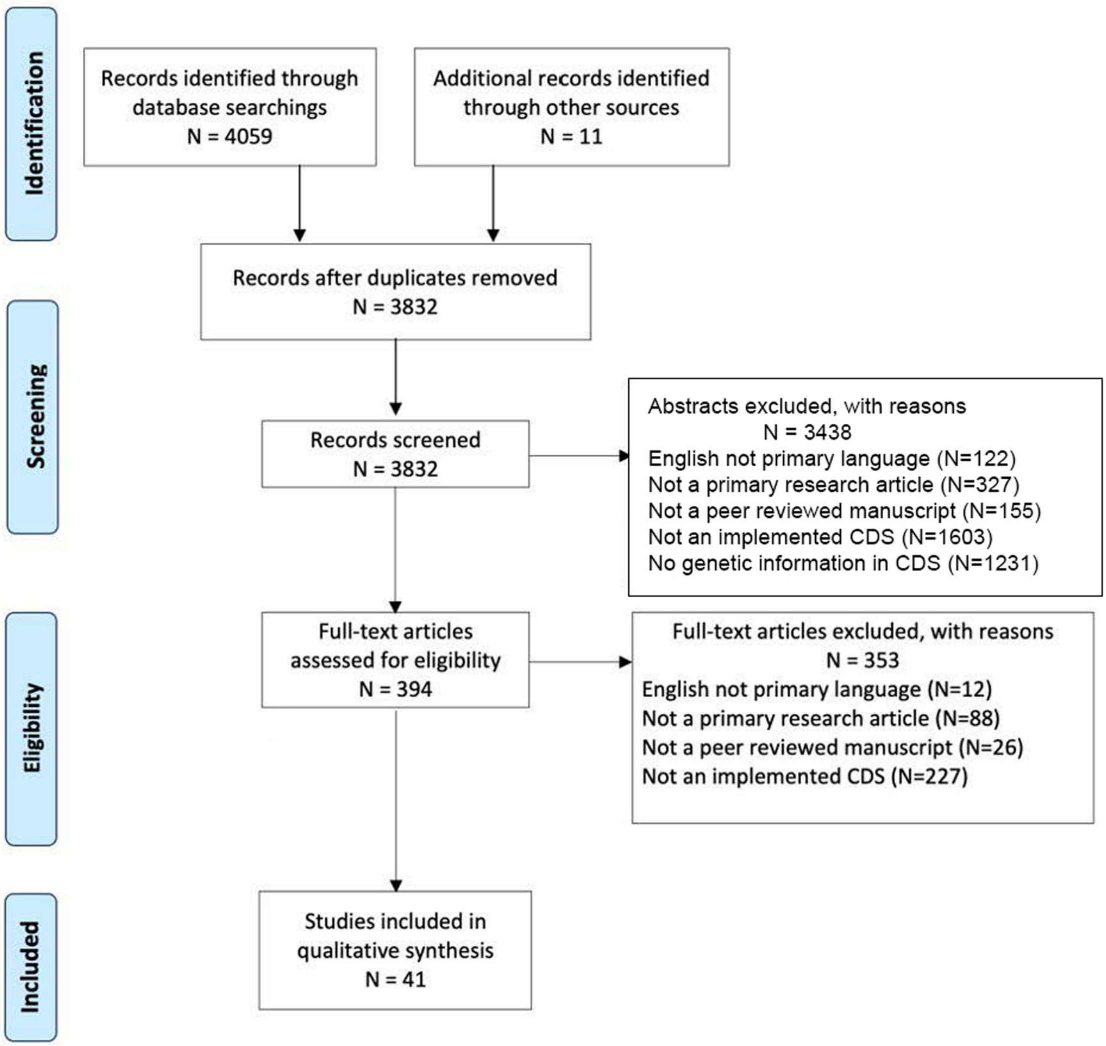


**Figure 1.** PRISMA diagram.

interoperability standards for CDS or genomic data storage and transfer (25 articles).

- *EHR standards-based:* A CDS tool that utilizes the native EHR and transfers genetic data using standard genomic data storage and transfer methods (2 articles).

## CDS systems for genetically guided cancer management

Of the 41 articles summarized in Table 1, genetically guided cancer management was the focus of 13 articles. These included 4 manuscripts related to the MeTree system,[37–40] an FHx-based risk assessment which provides FHx-driven CDS for several conditions including cancer,[41] 6 manuscripts on other FHx-driven CDS tools for cancer management,[42–47] and single manuscripts on the following subjects: population health EHR FHx-driven CDS[48] and genetically matching clinical trial patients.[49]

## CDS for pharmacogenomics

CDS to support PGx implementation was the focus of 23 of the implemented CDS systems (Table 1). Large research groups (eMERGE, PREDICT, U-PGx) composed of several institutions were responsible for 3 studies.[59,61,65] Single site/organization implementation was described in 17 articles[49–58,60,63,64,67–71] and clinical workflow and information presentation were highlighted as both barriers and facilitators. The remaining papers included implementations with a focus on the role of pharmacists,[62] experiences learned from transitioning from single-gene testing to panel-based testing,[72] and experiences of providers with the implemented systems.[66]

## Other CDS systems for genetically guided precision medicine

Five primary research articles focused on other clinical domains including 2 condition-specific tools designed within an institution: one directed toward anesthesiologists to help with management of patients at risk for malignant hyperthermia[74] and another for autism diagnosis and treatment.[73] A third study described a pregnancy genomic family planning tool using FHx as a driver for the CDS.[75] A fourth study using a pregnancy support tool to manage prenatal genetic risks using FHx.[76] Last, a study by Marwaha et al. described a tool using facial phenotyping to assist with diagnosis of genetic syndromes.[77]

## CDS type and critical features

Point of care alerts (32 out of 41) were the most common mechanism to deliver interventions, used in the gCDS systems

**Table 1.** Summary of primary research on CDS systems for genetically guided precision medicine.

| Citation and name of system (if applicable) | Manuscript summary and trial details (if applicable) | Users and study location | CDS purpose and clinical focus |
|---|---|---|---|
| Genetically guided cancer management | | | |
| Orlando, 2013; MeTree[37] | System description and initial implementation of MeTree, a family health history-based risk assessment tool which found MeTree can be integrated into PCP's workflow to improve adherence to guidelines | Patients and clinicians in USA | Decision support for diseases that have a strong impact on population health |
| Wu, 2013; MeTree[38] | Implementation review of MeTree at 3 PCP clinics found broad acceptance from patients and providers | Patients and clinicians in USA | Decision support for diseases that have a strong impact on population health |
| Wu, 2019; MeTree[39] | RE-AIM evaluation of MeTree, which found the system can be effectively implemented in diverse health systems | Patients and clinicians in USA | Collect information from patients about FHx and utilize information with PCPs |
| Wu, 2022; MeTree[40] | MeTree (a FHx web facing tool) increased post-intervention discussion between patients and providers, and increased uptake of risk recommendations | Patients and clinicians in USA | FHx collection tool that delivers CDS in PDF format to the EHR |
| Rubinstein, 2011; Family Healthware[41] | System description and implementation review for Family Healthware, which prompts PCPs about risk factor for 6 common conditions found no effect of FHx prevention messages on cancer screening | PCPs in USA | Collect FHx for 6 common diseases, stratifies risks, and provides tailored prevention messages |
| Zazove, 2015; Family Healthware[42] | Implementation review of Family Healthware, found no change in identification or screening of patients | PCPs in USA | Prompt PCPs about family history risk |
| Cunningham, 2012; BOADICEA[43] | System description and implementation review for BOADICEA, a web tool used to assess risks to patients with FHx of breast and ovarian cancer | PCPs in UK | Assess risks to patients with a family history of breast and ovarian cancer |
| Del Fiol, 2020; GARDE[44] | System description and implementation review for CDS GARDE platform, which searches the EHR to identify candidates for genetic testing of hereditary cancer syndromes based on FHx | Genetic counselors in USA | Use FHx to identify candidates for genetic evaluation of hereditary cancer syndromes |
| Lemke, 2020; Genetic Wellness Assessment (GWA)[45] | System description and implementation review for GWA found improved clinical care and several barriers to implementation | Patients and clinicians in USA | Patient administered FHx screening tool presented FHx to clinicians |
| Wurtmann, 2022; USA[46] | EHR alerts were linked to a genetic cancer screening tool which identifies individuals with hereditary breast and ovarian cancer related to *BRCA1/2*, Lynch syndrome, and other diseases due to personal and FHx | Patients and clinicians in USA | Refer patients at high risk for hereditary cancer to a genetic counselor |
| Doerr, 2014; MyFamily; USA[47] | Implementation Review for MyFamily FHx tool found improvement on quality and consistency of clinical care | Patients and clinicians in USA | FHx collection tool that delivers CDS through the EHR at the point of care |
| Yin, 2021; Ask2Me.org; USA[48] | Implementation review of Ask2Me.org which provides evidence-based risk predictions for individuals with pathogenic variants | Clinicians in USA | To provide evidence-based risk predictions for individuals with pathogenic variants in cancer genes |
| Zeng, 2019; OCTANE; USA[49] | System description and implementation review for OCTANE, an oncology clinical trial annotation engine designed to create a database for patient-trial matching | Clinical trial annotators in USA | To match patients with a relevant clinical trial |
| CDS for Pharmacogenomics (PGx) | | | |
| Gill, P.S.; 2021; Arkansas[50] | System description of Epic's genomic indicators to return results for a clinical PGx test | Clinicians in USA | To provide an information alert to clinicians about potential PGx drug interactions |
| Manzi, S.F.; 2016; Boston[51] | System description and implementation review for system which prompts clinicians based on laboratory results of PGx test found improved clinical outcomes | Clinicians in USA | To provide information about PGx to clinician at time of patient interaction at time of order |
| Hinerer, M; 2017; Germany[52] | Clinical surveys of sites with PGx implementation reviewed barriers and facilitators for implementing PGx | Clinicians in Germany | PGx CDS evaluation at 8 German hospitals |
| Zastrozhin, M; 2020; Russia[53] | Implementation of www.pgx2.com as a PGx interpretation tool found improved dosing and reduced side effects | Clinicians in Russia | Interpret and display results of PGx testing |
| Marrero, R.; 2020; GatorPGx[54] | Implementation review of GatorPGx which provides recommendations for PGx implementation | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Aquilante, C.; 2020; Colorado[55] | Implementation review for biobank PGx returning results preemptively to institutions | Clinicians and patients in USA | Preemptive PGx testing alerts clinicians at point of care |

(continued)

**Table 1.** (continued)

| Citation and name of system (if applicable) | Manuscript summary and trial details (if applicable) | Users and study location | CDS purpose and clinical focus |
|---|---|---|---|
| Borden, B.; 2018; GPS; USA[56] | Implementation review for Genomic Prescribing System PGx results found implementation barriers and increased utilization of PGx information | Clinicians in USA | Delivering PGx alerts to clinicians |
| Hernandez, W.; 2020; GPS; USA[57] | Implementation review for GPS PGx found genetic ancestry specific care improvements | Clinicians in USA | Using genetic ancestry to enable preemptive PGx alerts |
| Petry, N.; 2019; South Dakota[58] | System description and implementation review of Epic BPA's for PGx information delivery found improved care with barriers to implementation | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Blagec, K.; 2018; Austria[59] | Implementation review of U-PGx GIMS which manages genetic information and communicates results found improved care through several complementary methods (paper, electronic, and mobile) | Clinicians in Austria | Enable PGx CDS by secure transfer of PGx test results with dosing recommendations, generation of a PGx report, and a safety card enabling mobile based PGx CDS |
| Danahey, K.; 2017; Illinois[60] | System description and initial implementation of Genomic Prescribing System (GPS) found improved information delivery for PGx information to clinicians | Clinicians in USA | An online, secure, electronic custom interface for storing and delivering point-of-care genomic information |
| Rasmussen-Torvik, L.; 2015; USA[61] | System description and initial implementation of eMERGE-PGx project for preemptive PGx in electronic health records | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Owusu-Obeng, A.; 2014; Florida[62] | Roles for pharmacists emerged during clinical implementation of genotype-guided clopidogrel therapy | Clinicians in USA | Genotype drives EHR alerts with patient-specific genotype-guided drug therapy recommendations |
| Bielinski, S.J.; 2014; Minnesota[63] | Preemptive PGx testing for biobank participants included targeted sequencing of 84 PGx genes. Synchronous real-time CDS is integrated in the EHR, flags drug–gene interactions and provides recommendations | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Goldspiel, B.R.; 2013; Maryland[64] | A PGx committee was established to develop CDS algorithms for medications with hypersensitivity reactions at the point of care. Prescribers overrode recommendations mainly due to previous tolerance | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Pulley, J.M.; 2012; Tennessee[65] | Operational implementation of prospective genotyping linked to an advanced CDS system. This included patient and clinician engagement | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Caraballo, P.; 2017; Minnesota[66] | A comprehensive operational model can support PGx implementation. Institutions can use this model as a roadmap to support similar efforts | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Obeng, A.O.; 2016; New York[67] | Development of a strategy that accounts for the complexity of race and leverages EHRs for algorithm variables and deploying point-of-care dose recommendations | Clinicians in USA | Preemptive PGx testing alerts for warfarin clinicians at point of care |
| Ramsey, L.B.; 2017; Ohio[68] | Implementation of PGx at Cincinnati Children's Hospital: the development, launch, lessons learned, and updates to the system after 14 years | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Bell, G.C.; 2014; Tennessee[69] | Active CDS delivered through an EHR facilitates gene-based prescribing and other application of genomics to patient care. Interruptive CDS appropriately guided prescribing in 95% of patients for whom they were issued | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |
| Wick J.A.; 2014; Ohio[70] | Implementation of PGx across 30 primary care sites led to changes in recommended drugs 82% of the time, preventing adverse drug events and increasing quality of care | Clinicians in USA | CDS indicates which patients meet criteria for PGx testing |
| Peterson, J.F.; 2013; Tennessee[71] | The design of a locally developed EHR supporting PGx has a generalizable utility. Genetic data from panel-based genotyping were sequestered from the EHR until drug–gene interactions met evidentiary standards. | Clinicians in USA | Preemptive and indication triggered PGx testing alerts clinicians at point of care |
| Cicali, E.J.; 2019; Florida[72] | The University of Florida health precision medicine program led the clinical implementation of alerts for 6 gene–drug pairs. Lessons learned include prescriber education methods, clear concise guidance on genotype-based actions, and real time clinical results are key. | Clinicians in USA | Preemptive PGx testing alerts clinicians at point of care |

(continued)

**Table 1.** (continued)

| Citation and name of system (if applicable) | Manuscript summary and trial details (if applicable) | Users and study location | CDS purpose and clinical focus |
|---|---|---|---|
| Other CDS for precision medicine | | | |
| Way, H.; 2021; Australia[73] | Implementation review for IntellxxDNA, a tool used to evaluate genetic data and return a genomically targeted report based on specific SNP's, improved care for patients | Clinicians in Australia | Collect information about specific SNP's and return in a report to the clinician (genomically targeted intervention strategies including nutrients, supplements, and lifestyle modifications) |
| Baye, J.F.; 2020; South Dakota[74] | Malignant hyperthermia susceptibility: Utilization of genetic results in an electronic medical record to increase safety | Clinicians in USA | Alert system for malignant hyperthermia susceptibility |
| Edelman, E.A.; 2013; Maryland[75] | Evaluation of a novel electronic genetic screening and CDS tool in prenatal clinical settings | Clinicians and patients in USA | Computerized intervention for identification and management of prenatal genetic risks using FHx |
| Reumkens, 2020; Netherlands[76] | Clinical surveys of decision aid to support couples during reproductive decision-making found barriers and facilitators to implementation | Clinicians and patients in Netherlands | Online decision aid used to supplement counseling in a targeting group to make reproductive decisions |
| Marwaha, 2021REF; Face2Gene[77] | System description and implementation review for Face2Gene, which was designed to use facial phenotyping to suggest a potential diagnosis | Clinicians in Canada | Phenotype-driven prediction of genetic diagnosis based on facial features |

Abbreviations: BPA = Best Practice Advisory; CDS = clinical decision support; EHR = electronic health record; FHIR = Fast Healthcare Interoperability Resources; FHx and FHH = family health history; GPM = genetically guided precision medicine; HL7 = Health Level Seven International; PCP = primary care physician; RE-AIM = reach, effectiveness, adoption, implementation, and maintenance framework; SNP = single-nucleotide polymorphism.

(Table 2). Barriers to the implementation of alerts were described in 17 of the gCDS systems including alert fatigue, difficult integration with EHR, and lack of understanding of genetic information presented with the alert. Facilitators to the implementation of alerts were also mentioned in 14 of the gCDS systems including lack of interruption of clinical workflow, integration with EHR, and access to point-of-care information. There was little work reported on designing tools to work as expert systems (eg, diagnostic decision support) or for workflow support (eg, support in genetic test ordering, clinical documentation). Lack of genetic data standards was mentioned several times as a barrier to multidepartmental and multiinstitutional cooperation.

## Standards in genetic support systems

Of the 41 articles, only 3 reported using standards-based approaches for storing and transmitting genetic information: GARDE (FHx information using FHIR),[44] U-PGx (allowed information to be transmitted via several standards including FHIR and HL7),[59] and eMERGE (HL7 Version 2).[61] All other systems either used proprietary approaches within vendor-based EHRs (eg, Epic, Cerner) for data transfer, or created a unique local data repository external to the EHR as a part of the CDS architecture. About 27 systems were integrated with the EHR, 3 of which reported a standards-based approach. The remaining 14 systems were considered stand-alone CDS systems.

## User-centered design for genetic clinical decision support systems

Of the 30 user-centered design strategies from implementation experts,[78] there were only 2 studies (Table 3) that adopted over 50% of the strategies (ie, define target user, examine automatically generated data). Many studies did not report any category of user-centered design, potentially indicating a lack of understanding or use of user-centered design principles.

## Implementation framework

Of the 41 articles, 2 described the use of an implementation framework to develop, implement, and evaluate the CDS system (Table 4). Implementation frameworks used included RE-AIM[38–40] and PRISM.[55] MeTree utilized RE-AIM to evaluate uptake and implementation processes. Pre-implementation data collection strategies included site visits, staff surveys, and qualitative interviews to assess for readiness and identify barriers and facilitators. MeTree evaluated the population representation to measure the effects on all races within the community. Surveys and interviews to assess the adoption of MeTree showed positive responses to change commitment but low positive levels of change efficacy.[37–40] Implementation surveys helped identify needs for patients to understand how to access laboratory test results. Aquilante's biobank research utilized the PRISM methodology and praised the ease of use for the final CDS system, even though the process took several years of multistakeholder engagements, analysis of clinical workflows, and iterative designs.[55] Other articles described implemented CDS systems with no formal application of implementation frameworks for measuring outcomes related to implementation.

## Facilitators and barriers for genetic CDS implementation

The facilitators for gCDS systems include implementation of point of care alerts (29 articles), genetic data integration within EHR (23 articles), providing evidence-based and personalized recommendations for diagnosis (21 articles),

**Table 2.** Summary of CDS taxonomy,[35] CDS architecture, information used to drive CDS, and CDS critical features.[36]

| | Prevalence (%) |
|---|---|
| CDS type (taxonomy) | |
| Point of care alerts/reminders | 78.0 (32/41) |
| Medication dosing support | 58.5 (24/41) |
| Order facilitators | 51.2 (21/41) |
| Relevant information display | 48.7 (20/41) |
| Expert systems | 36.6 (15/41) |
| Workflow support | 12.2 (5/41) |
| CDS architecture type | |
| EHR proprietary | 61.0 (25/41) |
| Stand alone | 34.1 (14/41) |
| EHR standards based | 4.9 (2/41) |
| Information used for CDS | |
| Genotype | 68.3 (28/41) |
| Family history | 29.3 (12/41) |
| Phenotype | 2.4 (1/41) |
| CDS critical features for success (defined by K. Kawamoto[36]) | |
| Computer-based CDS | 100 (41/41) |
| Actionable recommendation provided | 92.7 (38/41) |
| Decision support provided in clinical workflow | 82.9 (34/41) |
| Decision support delivered at time and location of decision | 73.2 (30/41) |

A CDS could use multiple types of information for a CDS, for these implemented CDS, one specific type of information drove the decision support.

multidisciplinary teams developing the CDS (12 articles), patient-facing component of CDS (9 articles), guideline accessibility (6 articles), institutional support (5 articles), and physician champions (5 articles).

The barriers identified in the review include lack of standards for storing and transmitting genetic information (26 articles), lack of genetic data integration within the EHR (20 articles), lack of understanding of genetic information by the clinician (18 articles), alert fatigue (16 articles), cost to healthcare system with inadequate information or analysis on cost-benefit (15 articles), clinical workflow disruptions (14 articles), lack of access to genetic personnel (7 articles), and lack of patient engagement materials (4 articles).

### Trend analysis

The publication volume on CDS for GPM was relatively constant throughout the timeline of the review. The major foci of the literature in this domain have been on pharmacogenomics, FHx CDS, and CDS for cancer management. See Appendix S5 for more information.

## Discussion

As the field of genetically GPM continues to grow rapidly, there is a pressing need for more research on how to effectively harness CDS to integrate these discoveries into everyday clinical practice. For instance, even in the realm of FHx-driven CDS, which is arguably the most established area of research concerning CDS for GPM, implementations of FHx-driven CDS tools have primarily focused on hereditary cancer management. Genotype driven tools have primarily focused on PGx applications. This indicates that there is still a considerable knowledge gap in applying genotype and FHx data for GPM in many clinical specialties. Within this review, there were only 5 implementations not related to cancer or PGx.

**Table 3.** Summary of user centered design strategies for genetic CDS.[78]

| User-centered design strategy | Percentage (%) |
|---|---|
| Define target users and their needs | 85.4 (35/41) |
| Examine automatically generated data | 80.4 (33/41) |
| Engage in live prototyping | 39.0 (16/41) |
| Engage in cycles of rapid prototyping | 29.3 (12/41) |
| Apply task analysis to user behavior | 24.3 (10/41) |
| Engage in iterative development | 12.2 (5/41) |
| Use generative object-based techniques | 12.2 (5/41) |
| Conduct experience sampling | 12.2 (5/41) |
| Design in teams | 9.8 (4/41) |
| Conduct heuristic evaluation | 9.8 (4/41) |
| Conduct usability tests | 9.8 (4/41) |
| Build a user-centered organizational culture | 9.8 (4/41) |
| Conduct interviews about user perspectives | 7.3 (3/41) |
| Conduct interpretation sessions with stakeholders | 4.9 (2/41) |
| Conduct focus groups about user perspectives | 4.9 (2/41) |
| Conduct design charrette sessions with stakeholders | 4.9 (2/41) |
| Collect quantitative survey data on potential users | 4.9 (2/41) |
| Define workflows | 2.4 (1/41) |
| Conduct observational field visits | 2.4 (1/41) |
| Prepare and present user research reports | 2.4 (1/41) |
| Recruit potential users | 2.4 (1/41) |
| Use associative object-based techniques | 2.4 (1/41) |
| Use dialogic object-based techniques | 2.4 (1/41) |
| Conduct artifact analysis | 0 (0/41) |
| Conduct cocreation sessions | 0 (0/41) |
| Conduct competitive user experience research | 0 (0/41) |
| Develop a user research plan | 0 (0/41) |
| Develop experience models | 0 (0/41) |
| Develop personas and scenarios | 0 (0/41) |
| Apply process maps to system-level behavior | 0 (0/41) |

The topics of these implementations are not trivial, relating to pregnancy and family planning, autism, using phenotype information to predict genetic disease, and malignant hyperthermia risk. Given the limited literature available on any specific topic within this field, it is crucial to invest in further research to expand and deepen our understanding of what generalizable principles for gCDS can be utilized to facilitate implementation across a broad set of GPM domains.

Clinicians without formal training in genetics are usually the first to encounter patients with genetic conditions. gCDS can help clinicians recognize a genetic condition and initiate condition-specific management. gCDS systems can potentially improve and support the uptake of genetic services, but the impact on influencing clinical decisions has not been measured well. The results of our systematic review indicate that there is growing interest in implementing gCDS systems for GPM, as there were 38 total tools reviewed in the previous systematic review,[33] 20 of which were not implemented (18 unique tools). In the 12 years since that review, we assessed 41 implementation articles, with 36 unique tools implemented. Major challenges still need to be addressed including the lack of use of standards-based approaches to integrate CDS with EHR systems, limited clinical trials using a rigorous study design, and underuse of implementation frameworks and outcomes assessment in the evaluation of CDS implementations.

We observed a strong reliance on point-of-care alerts as the primary CDS type. Studies reported alerts as both facilitators and barriers to successful implementation. The characteristics that facilitate implementation include proper integration into the workflow (MeTree,[37–40] GatorPGx,[54] Genetic Wellness Assessment[45]) and minimization of interruptions (MeTree).

**Table 4.** Summary of implementation frameworks utilized for genetic clinical decision support systems.

| Citation and name of system (if applicable) | Manuscript summary and trial details (if applicable) | Implementation framework used |
|---|---|---|
| Orlando; 2013[37]; MeTree | System description and initial implementation of MeTree, a family health history-based risk assessment tool which found MeTree can be integrated into PCP's workflow to improve adherence to guidelines | RE-AIM: reach, adoption, implementation, maintenance |
| Aquilante, C.[55]; 2020; Colorado | Implementation review for biobank PGx returning results preemptively to institutions | PRISM: practical, robust implementation and sustainability model |

Interruptive alerts are often viewed as a barrier, whereas those that seamlessly integrate into the workflow act as facilitators.[41,42] The confusion of alerts being both a facilitator and barrier highlights the need to design alerts according to the user needs. The lack of more sophisticated CDS, such as expert systems and CDS that provide workflow support, speaks to the difficulty in creating an effective multidisciplinary team which would have the knowledge necessary to design the systems. However, the high rate of adoption of gCDS tools with features previously identified by Kawamoto et al.[36] as success factors for CDS (Table 2) suggests that CDS designers may be following lessons learned from prior research.

The lack of user-centered design reported in the articles is noteworthy given the necessity of user input and feedback throughout the iterative process of creating a successful gCDS system. Our review found that few gCDS tools evaluate workflow integration through clinician input to the process, or conducted usability tests. Future studies should follow user-centered design approaches in the design and implementation of gCDS. This recommendation was reached independently by participants in the Genomic Medicine XIII meeting (GMXIII) that proposed a research agenda to support the development and implementation of genomics-based clinical informatics tools and resources.[79] One of the meeting's short-term research priorities was "Developing user-friendly clinician- and patient-centered genomics-based tools and workflows". By emphasizing strategies that have a more positive impact, such as integrating alerts and reminders into the workflow and utilizing user-centered design principles, the effectiveness of CDS tools can be enhanced ultimately improving patient outcomes in GPM.

User-centered design can also aid integration, allowing gCDS tools to be compatible with existing health information systems, such as EHRs, laboratory information systems, and fit within the clinical workflow. User-centered design principles can help to establish the specific needs for data exchange and system interfaces, enabling seamless integration and promoting the effective use of genomic data across the healthcare ecosystem. In addition, these design principles promote evidence-based practice by encouraging the development of gCDS systems based on the best available evidence.

Implementation frameworks can also play a role in the development of CDS systems for GPM. These frameworks provide a structured approach that guides the design, integration, and evaluation of gCDS systems within the healthcare setting. Studies that used an implementation framework[37,40,55] facilitated stakeholder engagement by identifying and engaging key stakeholders, including healthcare providers, patients, administrators, and IT professionals, in the development and adoption of gCDS tools. This collaborative approach ensures that the system meets the needs of all users and promotes its acceptance and utilization. gCDS systems which didn't use implementation frameworks may have resolved specific barriers mentioned in the gCDS systems with their use, but this could not be assessed from the published results.

Implementation frameworks also facilitate evaluation and continuous improvement by providing a roadmap for assessing the impact of gCDS tools on clinical practice, patient outcomes, and healthcare costs. This evaluation process enables ongoing refinement and improvement of the system, ensuring that it continues to meet the evolving needs of clinicians and patients. The need for evaluation of the implementation of gCDS and other information systems to support GPM was endorsed in both the short and long-term research recommended by GMXIII.[79]

Our review revealed that the lack of use of standards for genomic data integration and representation remains a barrier to the incorporation of genetic information into EHRs and gCDS systems. Despite the previously highlighted importance of adopting standards[26,27] including prioritization in the recommendations from GMXIII, they are not being widely incorporated or are not being reported in the articles. Data standardization is mentioned in nearly all of the implementations, yet only 3 mentioned a specific data standard. Several articles mentioned the cost and resource constraints associated with implementing standards could deter smaller institutions from adopting them.[48,52,55,59] The issue of resource constraints affecting equitable implementation of gCDS and other information systems was highlighted in the recommendations of GMXIII. Implementations like the GatorPGx[54] and Family Healthware[41,42] call for standards as a way to promote data accuracy and consistency. Multi-institution projects (U-PGx, PREDICT, eMERGE) also cited data standards as a way to enable the scalability of gCDS systems, allowing them to be more easily implemented in diverse healthcare settings.

## Strengths and limitations

Use of a validated and systematic approach to the literature review was the major strength of this study. While the conclusions are necessarily subjective, they correspond to recommendations for the research agenda independently developed through an expert consensus process at the GMXIII meeting. In addition to the mentions in the text, Figure 2 maps the conclusions from the review to the recommendations from GMXIII.

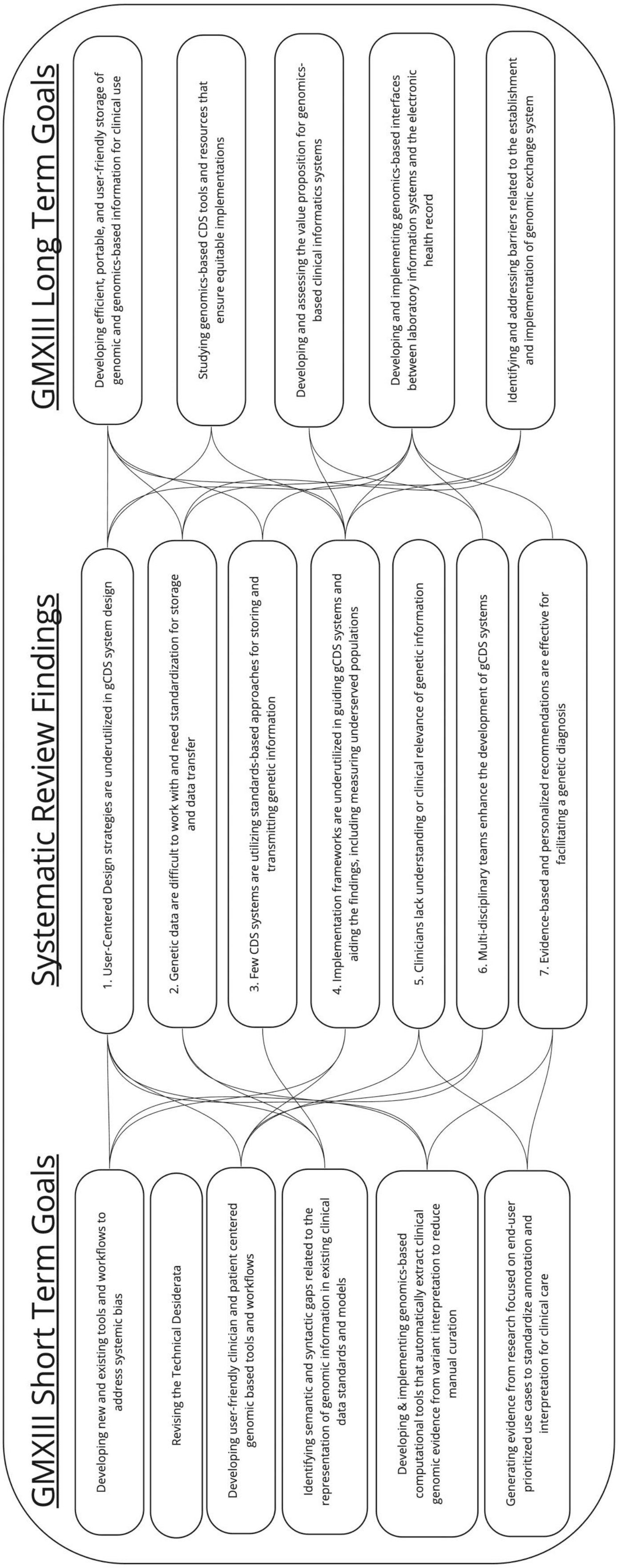


**Figure 2.** GMXIII recommendations, systematic review mapping results to recommendations.

In terms of limitations, this study does not provide a quantitative meta-analysis of the impact of CDS interventions for GPM. A meta-analysis was not possible due to the limited number of outcome studies in this field and the heterogeneous nature of the various interventions and clinical domains. Second, we only included manuscripts written in English, which may have led to some relevant manuscripts being excluded. Third, we only looked at published manuscripts, which excluded any tools that were developed for the commercial market and have been implemented but may not have published results of any evaluations that performed on these systems.

## Conclusion

Genetic disorders affect a significant portion of the population and present challenges for healthcare providers who may lack the necessary proficiency or confidence to effectively utilize genetic testing in clinical care. As genomic CDS tools advance and the cost of genetic testing decreases, GPM offers the potential to tailor medical treatments to individual patient characteristics. However, barriers persist in integrating genomic data into EHRs and CDS systems, which are critical for managing genetic testing and optimizing patient outcomes. Our systematic review of gCDS systems demonstrates that more systems are being implemented in healthcare systems, but also highlights the need for improvements in the integration of standard genomic data and the evaluation of these systems to maximize their potential in clinical practice.

To address these challenges, future research is needed to explore the use of user-centered design, unified standards-based approaches for developing gCDS tools, and implementation frameworks in the design, implementation, and evaluation of these tools. By promoting stakeholder engagement, data management, interoperability, evidence-based practice, and continuous improvement,[37–40,55] implementation frameworks can facilitate the successful adoption and utilization of genomic CDS tools in healthcare settings ultimately ensuring healthcare professionals will be better equipped to make informed decisions and improve patient care by incorporating the latest advancements in genomics into their practice.

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## Author contributions

D.J., G.D.F, K.M.R., and M.S.W. conceived and planned the review. D.J., N.S., G.I., E.J., and M.S.W. carried out the abstract and full-text review. D.J., M.S.W., K.M.R., K.K., and G.D.F. contributed to the interpretation of the results. D. J. took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis, and manuscript.

## Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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## Conflict of interest

None declared.

## Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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